GAURIKA DIWAN

Software Engineer | Backend & Cloud | AI & Automation

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Full-Time Australian Working Rights (Subclass 485)

PROFESSIONAL SUMMARY

Software Engineering (Honours) graduate from the University of Queensland, now a Graduate Engineer at Reliance Jio building Python-based MCP servers, API integrations and network automation for national-scale 5G infrastructure. Hands-on with FastAPI, FastMCP, Azure DevOps, Docker, AWS and CI/CD, with delivered results including a 40% reduction in provisioning effort and 91% model accuracy on 50,000+ records. Seeking Software Engineering roles across backend, cloud, platform and AI-enabled systems.

TECHNICAL SKILLS

Cloud & DevOps: Microsoft Azure, Azure DevOps, AWS (ECS, Lambda), Docker, Docker Compose, CI/CD, GitHub Actions, Linux, Git

Backend & APIs: Python, FastAPI, Flask, Pydantic, SQLAlchemy, REST APIs, HTTP APIs, WebSockets, YAML / API specifications

AI & Agentic Systems: MCP servers (FastMCP), A2A workflows, LangChain, Ollama, FAISS, PyTorch, pandas

Security & Integration: JWT / JWKS authentication, centralised token management, service-to-service authentication

Data & Languages: PostgreSQL, SQL, JavaScript, React, Java, pytest, time-series forecasting

PROFESSIONAL EXPERIENCE

Reliance Jio | Graduate Engineer – Software & Network Infrastructure *Apr 2026 – Present*

- Develop Python-based MCP servers and API integrations that expose network operations and enterprise workflows as structured, tool-callable capabilities using FastAPI, Pydantic and FastMCP.
- Build and validate MCP-based workflow automation capabilities, supporting 7 lifecycle tools for structured event management, updates and associated file operations within an enterprise integration environment.
- Enhance an MCP server management platform with server registration, tool discovery, API specification-driven code generation, execution monitoring and validation workflows.
- Work on centralised token management and secure service-to-service communication, including JWT/JWKS-based authentication and integration troubleshooting across MCP services.
- Develop and test integrations between enterprise systems and automation services, enabling programmatic notifications, messaging, workflow status updates and API-driven operations using Python, HTTP APIs, YAML, Git and Azure DevOps.

The University of Queensland | Undergraduate Researcher *Jul 2024 – Jul 2025*

- Developed Python forecasting workflows on 10+ years of Queensland energy and weather data to analyse temporal patterns and support predictive modelling.
- Improved forecasting accuracy by 15% through experimentation, feature analysis and model refinement across multiple forecasting approaches.

Reliance Jio Brain | ML & Software Development Intern *Dec 2024 – Feb 2025*

- Developed and evaluated machine learning models on 50,000+ records, achieving 91% accuracy on the target prediction task.
- Built a Flask REST API exposing model inference for application integration, optimised to deliver sub-second response times.

L2X Brazil | Network Automation Intern, Cisco Sydney Engagement *Dec 2023 – Feb 2024*

- Developed Python- and Docker-based network automation workflows to streamline network provisioning and configuration.
- Reduced provisioning effort by 40% and configuration defects by 15% through automation and improved execution consistency.

PROJECTS

MoviePicker — Real-Time Web Application *React, Flask, PostgreSQL, WebSockets, AWS ECS* *Feb 2025 – May 2025*

- Built React user flows integrated with a Flask REST API in a team project, delivering real-time match notifications.
- Co-authored architecture decision records for a modular, containerised backend deployed on AWS ECS and Lambda.

Alzheimer's Disease Classification *Python, PyTorch, torchvision, Matplotlib* *Sep 2023 – Oct 2023*

- Built a Vision Transformer from scratch to classify brain MRI slices, reaching 95% accuracy on ADNI data through preprocessing and augmentation.
- Validated model robustness against medical imaging metrics to ensure reliable diagnostic classification.

EDUCATION

Bachelor of Engineering (Honours), Software Engineering — Minor in Data Science *Graduated 2025*

The University of Queensland, Brisbane, Australia | GPA 5.5/7

- Women in Engineering Student Leader, UQ